Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Part Test - 1_P1 [2023 P1]

Test Date: 10th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCE-2023-P1-Model

Time: 3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 3)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 4 – 7)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 8 – 13)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 20)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 21 – 24)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 25 – 30)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 37)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 38 – 41)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 42 – 47)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

MATHEMATICS

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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01. Let $f:R\to R$ be a differentiable function satisfying $f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2}\forall x,y\in R$.

If $f(0)=1$ and $f'(0)=-1$, then which of the following is(are) correct?

A) $f\left(|x|\right)$ is discontinuous at one point

B) number of solution(s) of the equation $f(x)=f^{-1}(x)$ is exactly one

C) $\sum_{r=0}^{10}(f(r))^2=286$

D) $\tan^{-1}(f(x))$ is derivable $\forall x\in R$

02. If A and B are square matrices of order '3' such that $A\,adj(2B)=16I_3$ and $\det(B)=2$, then which of the following option(s) is/are true (where $adj(X)$ denotes adjoint of matrix (X), X^{-1} denotes inverse of matrix (X) and det(X) denotes determinant value of matrix(X))

A) $(A^{-1}(adjB))^{-1}=4A^2$ **B)** $(A^{-1}(adjB))^{-1}=B^2$

C) $B(adj\,A)=8I_3$ **D)** $B(adj\,A)=4I_3$

03. Let $f:A\to B$ and $g:B\to C$ be two functions and $gof:A\to C$ is defined. Then which of the following statement(s) is/are INCORRECT?

A) If *gof* is into, then *f* must be into
B) If *f* is into and *g* is onto, then *gof* must be onto function
C) If *gof* is one-one, then *g* must be one-one
D) If *f* is surjective and *g* is injective, then *gof* must be surjective

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If **ONLY** the correct option is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: −1 In all other cases

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04. The number of solution(s) of the equation $\left|2\sin x - \text{cosec}x\right| = 1$ in the interval $(0, \pi)$ is/are

A) 3 **B)** 4 **C)** 5 **D)** 6

05. Let $P_n = P_{n-1} + P_{n-2}\ \forall\ n \geq 2$ and $n \in N$, where $P_0 = 0$ and $P_1 = 1$, then which of the following is/are true?

A) $\sum_{k=1}^{\infty} \frac{P_k}{3^k} = \frac{3}{5}$ **B)** $\sum_{k=1}^{\infty} \frac{P_k}{3^k} = \frac{2}{5}$ **C)** $P_9 = 36$ **D)** $P_9 = 55$

06. If $f(\theta) = \dfrac{\sin 7\theta + 6\sin 5\theta + 17\sin 3\theta + 12\sin\theta}{\sin 6\theta + 5\sin 4\theta + 12\sin 2\theta}$, then which of the following options is correct?

A) $f\left(\frac{\pi}{3}\right) = \frac{1}{2}$ **B)** $f\left(\frac{\pi}{5}\right) = \frac{1+\sqrt{5}}{4}$

C) $f\left(\frac{2\pi}{5}\right) = \frac{\sqrt{5}-1}{4}$ **D)** $f\left(\frac{2\pi}{5}\right) = \frac{\sqrt{5}-1}{2}$

07. A line with direction cosines proportional to 2, 1, 2 meet the lines $x = y + 2 = z$ and $x + 2 = 2y = 2z$ at P and Q respectively. The distance between P and Q is

A) 2 **B)** 3 **C)** 6 **D)** 8

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks: +4 If ONLY the correct integer is entered; Zero Marks: 0 In all other cases.

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08. The number of integral values of x satisfying the equation $\left|x^4.3^{|x-2|}.2^{|x-5|}.5^{x-1}\right| = -x^4.3^{|x-5|}.2^{|x-2|}.5^{x-1}$.

09. Consider $f(x)=\dfrac{(x^2-1)(x^2-4)}{g(x)}, x \in R-\{a,b\}$, where $g(x)$ is a polynomial of degree ≤ 4.

$x=a,b$ are not in the domain of $f(x)$ but limit at these points exist.

The following three information are also known

(i) $\lim\limits_{x\to\pm\infty} f(x) = +\infty$

(ii) $\lim\limits_{x\to -1} f(x) = 6$

(iii) $\lim\limits_{x\to -2} f(x) =$ a non-zero finite number

Let L and M denotes the number of points of discontinuity and the number of point of non-differentiability of $y=|f(x)|$ respectively in $(0,\infty)$ then the value of (L+M) is equal to_____

10. If $f(x)=\sin^{-1}(x+2019)-\cos^{-1}(x+2021)$ and $g(x)=\sin^{-1}(x+2017)+\cos^{-1}(x+2019)$ and x_1, x_2 lies in domian of $f(x)$ & $g(x)$ respectively, then

$\left|\lim\limits_{x\to x_1+x_2} \dfrac{\sin(x-2x_1-2)-\sin(2x-3x_2+2022)}{\tan(x-x_1-x_2)}\right|$ is equal to_____

11. If $\vec{V}_1=\hat{i}+\hat{j}+\hat{k}; \vec{V}_2=a\hat{i}+b\hat{j}+c\hat{k}$ where $a,b,c\in\{-2,-1,0,1,2\}$, then the number of non-zero vectors $\vec{V}_2$ which are perpendicular to $\vec{V}_1$ is equal to_____

12. Let $f(x)=x^2+bx+c\ \forall x\in R,(b,c\in R)$ attains its least value at $x=-1$ and the graph of $f(x)$ cuts y-axis at $y=2$ .Then the least value of $f(x)\forall x\in R$ is:

13. If system of linear equations $(a-1)x+z=\alpha, x+(b-1)y=\beta$ and $y+(c-1)z=\lambda$ (where $a,b,c\in Z$) does not have a unique solution, then maximum possible value of $|a+b+c|$ is

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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14. Let (x_1,y_1,z_1) and (x_2,y_2,z_2) are 2 set of solutions satisfying the following system of equations

$\log_{10}(2xy)=4+(\log_{10}x-1)(\log_{10}y-2)$

$\log_{10}(2yz) = 4 + (\log_{10} y - 2)(\log_{10} z - 1)$

$\log_{10}(zx) = 2 + (\log_{10} z - 1)(\log_{10} x - 1)$ such that $x_1 > x_2$,

Then match List-I with List-II and select the correct answer using the code given below the list.

List – I		List – II	
P)	$\frac{y_1}{x_1}$	1)	2
Q)	$\frac{z_1}{x_2}$	2)	100
R)	$\frac{z_1 x_2}{z_2}$	3)	1000
S)	$\frac{y_2 + z_1}{x_2}$	4)	150

Codes:

A) $P \to 1; Q \to 2; R \to 4; S \to 3$ **B)** $P \to 1; Q \to 2; R \to 3; S \to 4$

C) $P \to 2; Q \to 1; R \to 4; S \to 3$ **D)** $P \to 2; Q \to 1; R \to 3; S \to 4$

15. Consider $P(x) = 2x^3 + x^2(3 - 7a) + x(3a^2 - 5a) + (3a^2 + 2a - 1)$.

Match List-I with List-II and select the correct answer using the codes given below the list.

List – I		List – II	
P)	Number of positive value(s) of 'a' for which $P(x) = 0$ has exactly two equal roots, is	1)	3
Q)	Number of positive value(s) of 'a' for which all roots of $P(x) = 0$ are negative, is	2)	1
R)	Absolute value of sum of integral values of 'a' for which one root of $P(x) = 0$ is greater than –1 and other root is less than –1, is	3)	2
S)	Number of integral values of 'a' for which all roots of $P(x) = 0$ lie in [–1, 2], is	4)	0

Codes:

A) $P \to 1; Q \to 4; R \to 2; S \to 3$ **B)** $P \to 2; Q \to 1; R \to 4; S \to 3$

C) $P \to 2; Q \to 4; R \to 1; S \to 3$ **D)** $P \to 4; Q \to 1; R \to 2; S \to 3$

16. Match List-I with List-II and select the correct answer using the codes given below the list.

List – I		List – II	
P)	Domain of the function $f(x) = \sin^{-1}[2x] + \tan^{-1}\frac{1}{[x]}$ is	1)	$\left[\frac{-1}{4}, \infty\right)$

Q)	If the domain of $f(x)$ is $[-2, 10]$, then the domain of $f(2x+4)$ is	2)	$\left[\frac{-1}{2}, 0\right)$
R)	Range of $f(x) = x^4 - x^2$ is	3)	$(0,1]$
S)	Range of $f(x) = \cos^2 x + \log_2\left(1 - \left[\sin^2 x\right]\right)$ is	4)	$[-3,3]$

(where [.] denotes greatest integer function)

Codes:

A) $P \to 2; Q \to 1; R \to 4; S \to 3$ **B)** $P \to 3; Q \to 4; R \to 1; S \to 2$

C) $P \to 2; Q \to 4; R \to 1; S \to 3$ **D)** $P \to 1; Q \to 4; R \to 3; S \to 2$

17. Match the following statement in List-I with the values in List-II

List – I		**List – II**	
P)	If the range of values of a such that a circle passes through the point of intersection of $y = x^2$ and $x^2 - y^2 = a^2$ is (α, β), then $\lvert\alpha - \beta\rvert$ equals	1)	24
Q)	The circle $x^2 + y^2 = 1$, cuts the x-axis at P and Q. Another circle with centre at 'Q' and variable radius 'r' intersects the first circle at R above the x-axis and the line segment PQ at S. If the area of ΔQRS is maximum when $r = \sqrt{\frac{M}{3}}$, then value of M is	2)	8
R)	Tangents drawn from (4, 0) to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ meet the circle at A and B. If the length of the chord AB is $\frac{6}{\sqrt{T}}$, then value of T is	3)	10
S)	If the distance between centres of two circles of radii 3, 4 is 25, then length of their transverse common tangent is	4)	1

Codes:

A) $P \to 1; Q \to 2; R \to 3; S \to 4$ **B)** $P \to 4; Q \to 3; R \to 1; S \to 2$

C) $P \to 4; Q \to 2; R \to 3; S \to 1$ **D)** $P \to 3; Q \to 1; R \to 4; S \to 1$

PHYSICS

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
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Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
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For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks..

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18. A thin copper tube of outer radius 0.5 cm carries a liquid flowing at $T = 100^{\circ}C$. The copper tube loses heat according to Newton's law with constant of proportionality $3\times10^{-3}\ cal/cm^2\sec^{\circ}C$. The temperature of surrounding is $20^{\circ}C$. Now we coat a layer of material with thermal conductivity $2.8\times10^{-3}\ cal/cm^{\circ}C\sec$. The layer is 0.5 cm thick. Assume that outer surface of layer losses heat with same constant of proportionality: (Take: $ln2 = 0.7$) Now

A) The rate of heat loss becomes $\frac{7}{8}$ times initial rate of loss of heat.

B) The rate of heat loss becomes $\frac{8}{7}$ times initial rate of loss heat.

C) The temperature T of outer surface of layer is approximately $65.7^{\circ}C$ in steady state.

D) The temperature T of outer surface of layer is approximately $42.2^{\circ}C$ in steady state.

19. Four particles A, B, C and D are kept at the vertices of a square of side l. Now all of them simultaneously start moving with same uniform speed ' v ' such that 'A' is moving towards 'B', B is moving towards 'C' and so on.

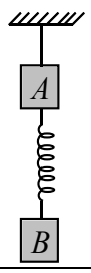
A) the time particles will take to meet each other is $\frac{a}{v}$

B) the time after the area of square formed by the particles becomes half is $\frac{\sqrt{2}-1}{\sqrt{2}}\frac{l}{v}$

C) the time particles will take to meet each other is $\frac{a}{\sqrt{2}v}$

D) the time after the area of square formed by the particles becomes half is $\frac{\sqrt{2}-1}{2\sqrt{2}}\frac{l}{v}$

20. In the given figure, both the blocks have equal mass and are in equilibrium and spring is mass less. Which of the following statement give correct description immediately after the thread joining block A to ceiling is cut?



A) Relative to the block A, acceleration of block B is 2g upwards.

B) Relative to the block B, acceleration of block A is 2g downwards.

C) Relative to the ground, accelerations of the block A and B are both g downwards.

D) Relative to the ground, accelerations of the block A and B are 2g downwards and zero respectively.

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

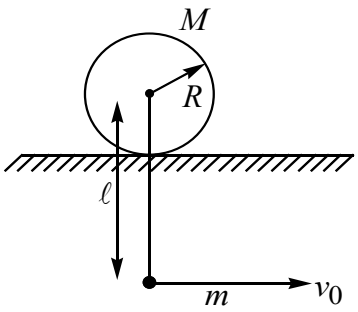
This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If **ONLY** the correct option is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: −1 In all other cases

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21. A disc of radius R and mass M is placed on smooth horizontal surface as shown in figure. A light rod of length ℓ is hung from the centre of disc and a small mass m is attached at the other end as shown in figure. Now a velocity V_0 is given to mass m. Find the maximum height which mass m can attain. (Assume rod cannot rotate more than 90°)

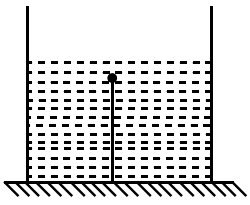


A) $\frac{1}{2}\frac{V_0^2}{g}\frac{m}{M+m}$ **B)** $\frac{1}{2}\frac{V_0^2}{g}\frac{M}{2m}$ **C)** $\frac{1}{2}\frac{V_0^2}{g}\frac{m}{2M}$ **D)** $\frac{1}{2}\frac{V_0^2}{g}\frac{M}{M+m}$

22. A particle is moving in a circle of radius R in such a way that at any instant the normal and tangential component of the acceleration are equal. If its speed at t = 0 is u_0, the time taken to complete the first revolution is

A) $\frac{R}{u_0}$ **B)** $u_0 R$ **C)** $\frac{R}{u_0}\left(1-e^{-2\pi}\right)$ **D)** $\frac{R}{u_0}e^{-2\pi}$

23. A large cylindrical vessel containing a liquid of density σ is kept on a horizontal surface. A small spherical object of density $\rho(<\sigma)$ is fastened to the bottom of the vessel with the help of a string as shown in the figure. Neglect the viscosity, and assume the gravity g vertically downwards. If the vessel is now given a constant horizontal acceleration equal to g towards right, then at the position of equilibrium of the spherical object with respect to the vessel, the deflection of the string with the vertical is:
(Assume height of water column is sufficiently large than length of string)



A) towards left with angle $\tan^{-1}\left(\frac{\sigma}{\sigma-\rho}\right)$.

B) towards right with angle $\pi/4$.

C) towards left with angle $\pi/4$.

D) towards right with angle $\frac{\rho}{\sigma-\rho}$.

24. 100 g of water is super cooled to $-10^{\circ}C$. At this point, due to some disturbance mechanized or otherwise, some of it suddenly freezes to ice. Final contents are

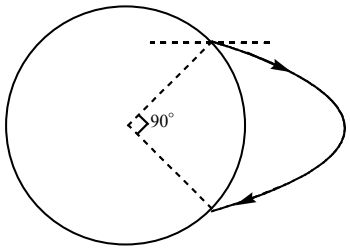
A) 12.5g ice + 87.5g water **B)** 50g water + 50g ice

C) 100g water **D)** 87.5g ice + 12.5 g water

SECTION-III
(INTEGER ANSWER TYPE)

• This section contains EIGHT (06) questions.
• The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
• For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
• Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If ONLY the correct integer is entered; Zero Marks: 0 In all other cases.

25. A particle is projected from the surface of the earth (of radius R_e and mass M_e) with speed equal to $\sqrt{\frac{GM_e}{R_e}}$,at certain angle from local horizontal as shown in the figure such that the angle subtended by arc between launching & landing site at earth's centre is $\beta = 90^{\circ}$. The maximum separation of the particle from the centre of the earth is $R_e\frac{\left(\sqrt{2}+x\right)}{\sqrt{2}}$. Find 4x? Consider earth to be uniformly dense and air resistance to be absent.

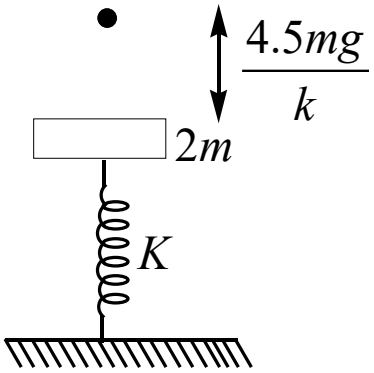


26. A solid has a volume v_0, when external pressure is p_0. Bulk modulus of the material is B. Now the pressure is increased to p. The additional energy per unit volume now stored in the materials is $\frac{3\left(p^2-p_0^2\right)}{zB}$. Find z

27. In the figure shown, a plate of mass 2m is at rest and in equilibrium. A particle of mass m is released from height $\frac{4.5mg}{k}$ from upper surface of the plate; where k is spring constant of the spring. The particle collides with the plate at time t = 0 and sticks to the plate. If

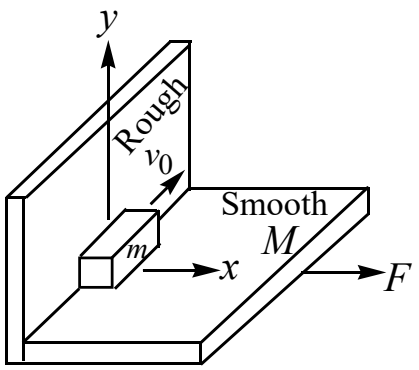
the time when the spring has maximum compression is $t = \frac{2\pi}{\lambda}\sqrt{\frac{3m}{k}}$ find the value of λ.

Neglect the duration of collision.

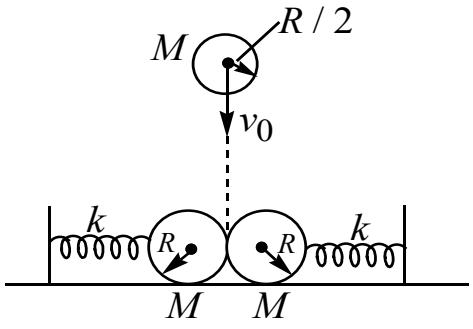


28. A container filled with air under pressure p_0 contains a soap bubble of radius R. The air pressure has been reduced to half isothermally and the new radius of the bubble becomes $\frac{5R}{4}$. If the surface tension of the soap water solution is S, p_0 is found to be $\frac{16nS}{R}$ SI unit. Find n.

29. Figure shows a block placed on a bracket. Bracket is placed on a smooth floor, it is pulled by a force F = 6N horizontally. Block is projected with velocity v_0 relative to bracket as shown in figure. Find time in second after which it stops relative to bracket. Horizontal surface of bracket is smooth while vertical surface is rough (Given: m = 1kg, M = 5kg, $v_0 = 4\,m/s$ $\mu = 0.5$), the motion of bracket is restricted in z-direction.



30. Two identical balls each of mass M (=1.25 kg) and radius R are kept in contact and are connected to two identical unstretched springs each of force constant k (=100 N/m). A third ball also of mass M but radius R/2 strikes with the two balls symmetrically with speed v_0 (=1.6 m/s) and comes to rest just after the impact. Find the maximum compression in each spring in cm if all the surfaces are smooth



SECTION – IV
(MATCHING TYPE)

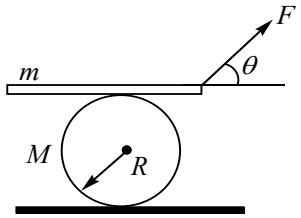
This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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31. Consider a solid cylinder of mass M and radius R lying on a rough horizontal plane. It has a plank lying on its top as shown in figure. A force F is applied on the plank such that the plank moves and causes the cylinder to roll. The plank always remains horizontal. There is no slipping at any point of contact. Match the quantities in Column-I with their values in Column-II.

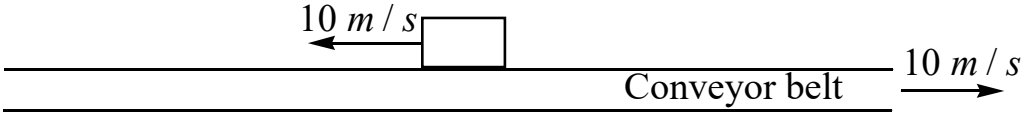


Column – I		Column – II	
A)	The acceleration of plank	P)	$\frac{3MF\cos\theta}{[3M+8m]}$
B)	The frictional force on plank	Q)	$\frac{MF\cos\theta}{[3M+8m]}$
C)	Acceleration of centre of mass of cylinder	R)	$\frac{8F\cos\theta}{[3M+8m]}$
D)	The frictional force on cylinder by the ground	S)	$\frac{4F\cos\theta}{[3M+8m]}$

A) $A \to R; B \to P; C \to S; D \to Q$ **B)** $A \to S; B \to P; C \to R; D \to Q$

C) $A \to S; B \to Q; C \to R; D \to P$ **D)** $A \to R; B \to Q; C \to S; D \to P$

32. A box of mass 1.0 kg moving with velocity 10 m/s towards the left is gently placed on a rough conveyor belt that is moving horizontally with a constant velocity 10 m/s towards the right. Coefficient of kinetic friction between the belt and the block is 0.5. Consider the time interval in which block slides on the belt. For this time interval, some physical quantities are mentioned in the first column and their possible value in SI units are mentioned in the second column. Suggest suitable match.



Immediately after the block is placed on the conveyor belt

Column – I		Column – II	
A)	Work of friction on the block with respect is ground	P)	Zero
B)	Magnitude of work of friction on the block with respect to the belt	Q)	20

C)	Distance slid by the block on the belt relative to the belt	R)	40
D)	Magnitude of change in momentum of block relative to ground	S)	200

A) $A \to P; B \to Q; C \to R; D \to S$ **B)** $A \to Q; B \to P; C \to S; D \to R$

C) $A \to P; B \to S; C \to R; D \to Q$ **D)** $A \to P; B \to S; C \to R; D \to S$

33. Match the temperature of a black body given in List-I with an appropriate statement in List-II, and choose the correct option.

[Given: Wien's constant as 2.9×10^{-3} m-K and $\frac{hc}{e} = 1.24 \times 10^{-6}$ V-m]

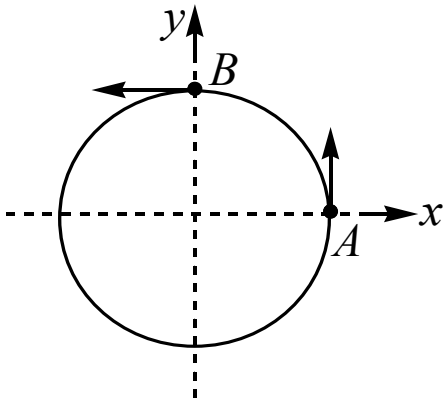
(Peak wavelength is wavelength corresponding to max spectral emissive power)

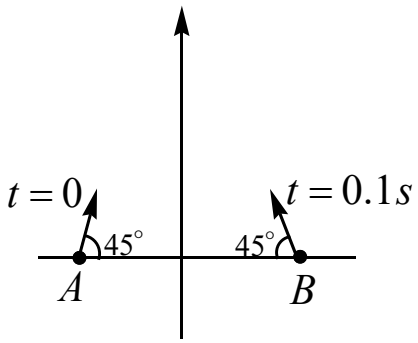
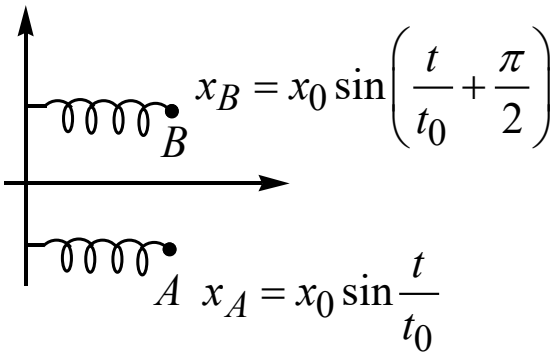
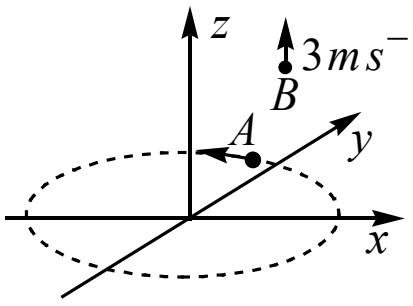
List – I		LIST – II	
P)	2000 K	1)	The radiation at peak wavelength can lead to emission of photoelectrons from a metal of work function 4 eV
Q)	3000 K	2)	The radiation at peak wavelength is visible to human eye.
R)	5000 K	3)	The radiation at peak wavelength will result in the widest central maximum of a single slit diffraction.
S)	10000 K	4)	The power emitted by a black body per unit a area is 1/16 of that emitted by a blackbody at temperature 6000K.s
		5)	The radiation at peak wavelength can be used to image humanbones.

A) $P \to 3, Q \to 5, R \to 2, S \to 3$ **B)** $P \to 3, Q \to 2, R \to 4, S \to 1$

C) $P \to 3, Q \to 4, R \to 2, S \to 1$ **D)** $P \to 1, Q \to 2, R \to 5, S \to 3$

34. List I describes four systems, each with two particles A and B in relative motion as shown in figure. List II gives magnitudes of their relative velocity (in ms^{-1}) at time $t = \frac{\pi}{3} s$.

List – I		List – II	
I)	A and B are moving on a horizontal circle of radius 1 m with uniform angular speed $\omega = 1\ rad\ s^{-1}$. The initial angular positions of A and B at time t = 0 are $\theta = 0$ and $\theta = \frac{\pi}{2}$ respectively. 	P)	$\frac{\sqrt{3}+1}{2}$
II)	Projectiles A and B are fired (in the same vertical plane) at t=0 and t = 0.1 s respectively, with the same speed	Q)	$\frac{\left(\sqrt{3}-1\right)}{\sqrt{2}}$

	$v = \frac{5\pi}{\sqrt{2}} m s^{-1}$ and at 45° from the horizontal plane. The initial separation between A and B is large enough so that they do not collide, $(g = 10 m s^{-2})$.  		
III)	Two harmonic oscillators A and B moving in the x direction according to $x_A = x_0 \sin\frac{t}{t_0}$ and $x_B = x_0 \sin\left(\frac{t}{t_0} + \frac{\pi}{2}\right)$ respectively, starting from t = 0. Take $x_0 = 1m$, $t_0 = 1$.  	R)	$\sqrt{10}$
IV)	Particle A is rotating in a horizontal circular path of radius 1m on the xy plane, with constant angular speed $\omega = 1\ rad\ s^{-1}$. Particle B is moving up at a constant speed $3\ m\ s^{-1}$ in the vertical direction as shown in the figure. (Ignore gravity.)  	S)	$\sqrt{2}$
		T)	$\sqrt{25\pi^2 + 1}$

Which one of the following options is correct?

A) $I \to R, II \to T, III \to P, IV \to S$ **B)** $I \to S, II \to P, III \to Q, IV \to R$

C) $I \to S, II \to T, III \to P, IV \to R$ **D)** $I \to T, II \to P, III \to R, IV \to S$

CHEMISTRY **Max Marks: 60**

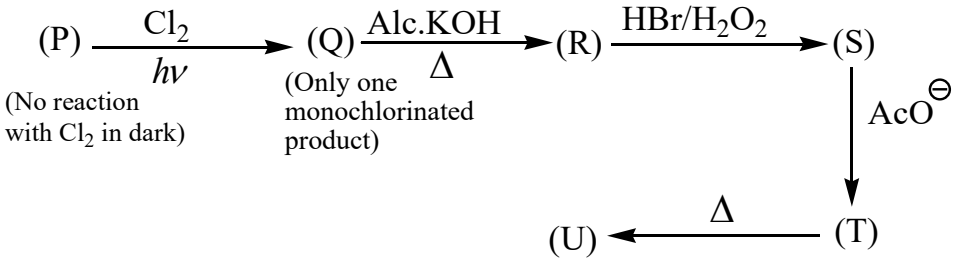
SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks..

35. Choose the correct option regarding following reaction sequence:



A) (R) will give same product with OMDM and HBO

B) E_i and E_2 involved in above sequence

C) Nucleophilic substitution reaction is involved

D) Free radical substitution and free radical addition reaction is involved.

36. Which of the following Quantities is/are zero at critical point-

A) Compressibility factor (Z) **B)** Molar Heat capacity

C) Surface Tension **D)** $\Delta H_{vaporisation}$

37. Which of the following statement(s) is/are correct?

A) H_2^+ has less bond dissociation enthalpy and more bond length than dihydrogen

B) Covalent radius of hydrogen (H) is less than that of fluorine(F) whereas ionic radius of hydrogen $\left(H^-\right)$ is more than that of fluorine $\left(F^-\right)$

C) Metallic radius of lithium $\left(Li\right)$ is less than that of magnesium $\left(Mg\right)$ whereas ionic radius of lithium $\left(Li^+\right)$ is more than that of magnesium $\left(Mg^{2+}\right)$

D) All d – block metals readily form metallic hydrides

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

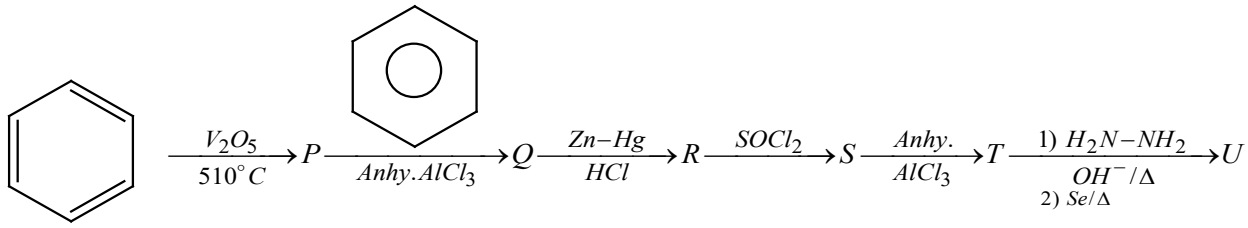
This section contains **FOUR (04)** questions.

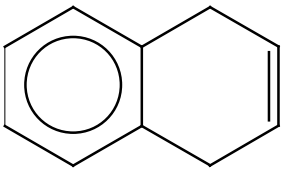
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

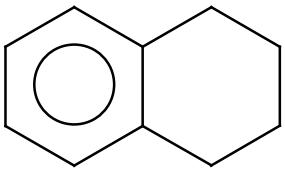
Full Marks: +3 If **ONLY** the correct option is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: −1 In all other cases

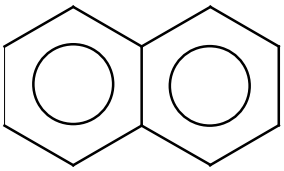
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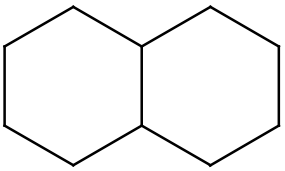
38. The final product 'U' is



A) 

B) 

C) 

D) 

39. A mixture of hydrocarbon C_2H_2, C_2H_4 & CH_4 in mole ratio of 2: 1: 2 is burnt completely in the presence of air containing 80% N_2 & 20% O_2 by volume. The mass of air required for the complete combustion of the one gm of mixture is

A) $\frac{1728}{112}$ **B)** $\frac{1528}{73}$ **C)** $\frac{1920}{120}$ **D)** $\frac{112}{1728}$

40. Which of the following statements is incorrect?

A) Li_2CO_3 and $LiNO_3$ decompose readily on heating to the oxide among the carbonates and nitrates of group 1 metals

B) Even though Na and Cu have same number of s – electrons in the outermost orbit, copper has more first ionization enthalpy than Na but Cu has less second ionization enthalpy than Na

C) Lattice energy (magnitude) of K_2O_2 is less than that of KO_2

D) Chlorides of both lithium and magnesium crystallises from their aqueous solutions as hydrated salts.

41. Which of the following statements is correct?

A) BF_3 is prepared by heating a mixture of calcium fluoride and boric oxide with concentrated sulphuric acid

B) Both BF_3 and BCl_3 are rapidly and completely hydrolysed by water

C) All four trihalides of the elements Al, Ga, In and Tl of the general formula $M^{III}X_3$ are stable ($X = F, Cl, Br, I$ and III : +3 oxidation state)

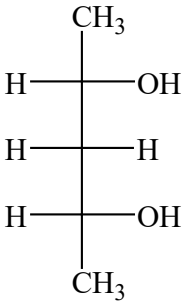
D) Extent of π – bonding among the three isoelectronic ions NO_3^-, CO_3^{2-} and BO_3^{3-} is $BO_3^{3-} > CO_3^{2-} > NO_3^-$

SECTION-III
(INTEGER ANSWER TYPE)

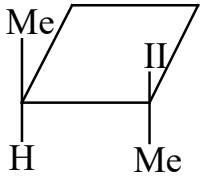
- This section contains EIGHT (06) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If ONLY the correct integer is entered; Zero Marks: 0 In all other cases.

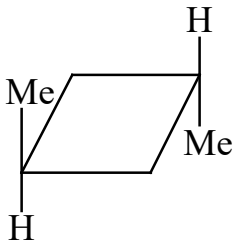
42. How many following compounds are optically inactive but can show optical isomerism?



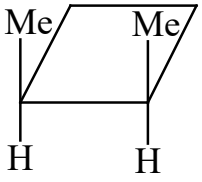
i)



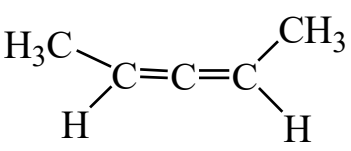
ii)



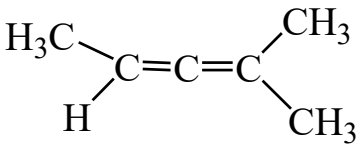
iii)



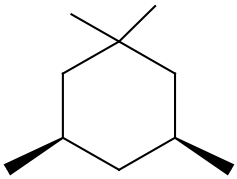
iv)



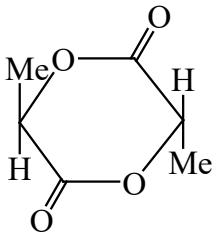
v)



vi)

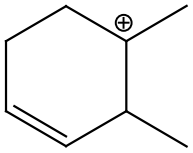


vii)

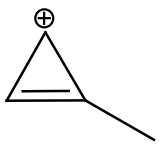


viii)

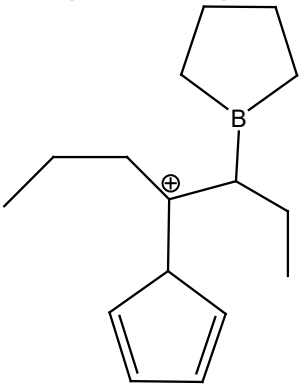
43. How many of the following species can undergo rearrangement?



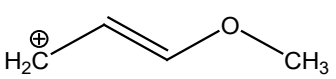
A)



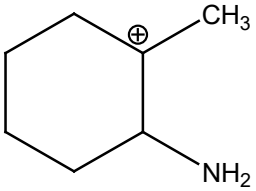
B)



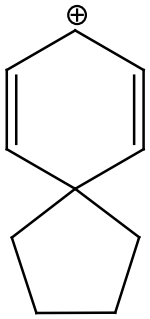
C) D)



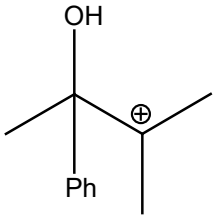
E)



F)



G)



44. For the reaction $Br_2(l) + Cl_2(g) \rightleftharpoons 2BrCl(g) K_p = 1 atm$. In a closed container of volume 164 litre initially 10 moles of Cl_2 are present at $27^0 C$. What minimum mole of $Br_2(l)$ must be introduced into the container, so that above equilibrium is maintained at a total pressure of 2.25 atm. Vapour pressure of $Br_2(l)$ at $27^0 C$ is 0.25 atm. Assume that volume occupied by the liquid is negligible.

$R = 0.0821$ lit atm/ mole K

Gram atomic mans of Br = 80 gram/mole

45. A gas consisting of rigid diatomic molecules was expanded in a polytropic process so that the rate of collisions of the molecules against the vessel's wall did not change. The molar heat capacity of the gas (in Cal/K-mol) in this process is

[Given R = 2 Cal/K-mol]

46. Give the atomic number of the inert gas atom in which the total number of d-electrons is equal to the difference in the number of total p- and s-electrons:

47. Consider the following reactions:

R–I: Number of moles of H_2O obtained during the reaction between H_2O_2 and Fe^{+2} (in acidic medium) is 'x'.
R–II: Number of moles of O_2 obtained during the reaction between H_2O_2 and MnO_4^- (in acidic medium) is 'y'.
R–III: Number of moles of O_2 obtained during the reaction between H_2O_2 and MnO_4^- (in basic medium) is 'z'.

Balance all the given three reactions and find the valve of (x+y+z)?

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>

Full Marks: +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

48.

Column-I (Pair of compounds)		**Column-II (Can be distinguished by)**	
A)	$CH_3-CH=CH_2$ and $CH_3-C\equiv CH$	P)	$NaNH_2$
B)	CH_3-CH_3 and $CH_3-C\equiv CH$	Q)	Br_2 / H_2O
C)	CH_3COOH and $CH_3-CH(OH)-CH_3$	R)	$Aq.\ NaHCO_3$
D)	CH_3-OH and $CH_3-CH=CH_2$	S)	Na metal
		T)	Aq. NaOH

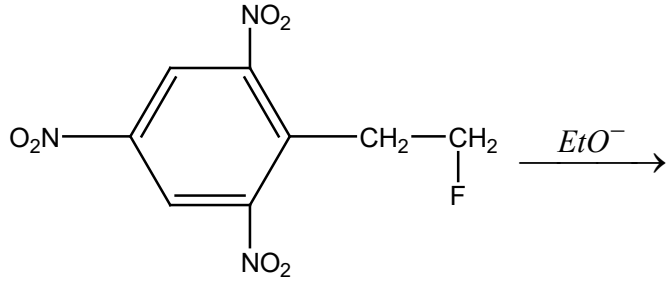
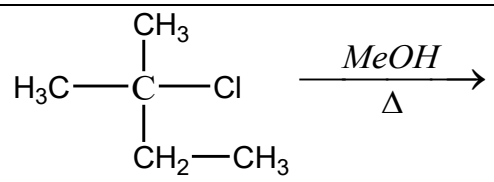
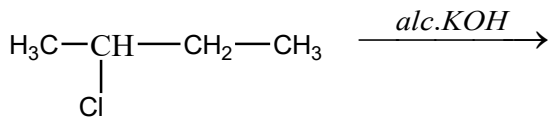
A) $A-P,\ B-PQS, C-RT, D-PQS$

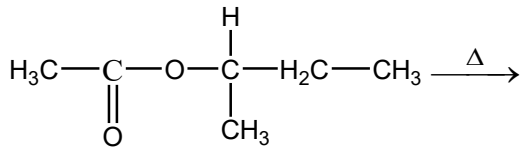
B) $A-PS,\ B-PQST, C-RT, D-PQS$

C) $A-PS,\ B-PQS, C-PRT, D-PQS$

D) $A-PS,\ B-PQS, C-RT, D-PQS$

49. Match the following.

List – I (Reaction)		**List – II (most appropriate mechanism expected)**	
P)		1)	E_2 mechnism
Q)		2)	Hoffman product is formed as major product
R)		3)	E_1CB mechanism

S)		4)	E_1 mechanism
		5)	It is single step reaction

A) P-1, 5; Q-1, 5; R-2, 3; S-2 **B)** P-3; Q-4; R-3; S-2

C) P-3; Q-4; R-1, 5; S-2,5 **D)** P-1, 5; Q-1, 2; R-1, 5; S-2

50. Match the column

Column-I		**Column-II**	
A)	Adiabatic expansion of an ideal gas against 1 atm	P)	$\Delta S_{sys} > 0$
B)	Condensation of vapour at normal boiling point equilibrium	Q)	$\Delta U = 0$
C)	Adiabatic reversible compression involving an ideal gas	R)	$\Delta S_{univ} = 0$
D)	Isothermal reversible expansion of an ideal gas	S)	w = 0
		T)	q =0

A) $A \to PT, B \to S, C \to R, T\ D \to P, Q, R$

B) $A \to PT, B \to R, C \to R, T\ D \to P, Q, R$

C) $A \to RT, B \to S, C \to PT, D \to P, Q, R$

D) $A \to RT, B \to R, C \to PT, D \to P, Q, R$

51. Match the molecules with the type of overlapping in their bridge bonds:

List – I		**List – II**	
P)	Be_2H_4	1)	$Sp^3 - S - Sp^3$
Q)	B_2H_6	2)	$Sp^2 - S - Sp^2$
R)	$Al_2(CH_3)_6$	3)	$Sp^2 - Sp^3 - Sp^2$
S)	$Be_2(CH_3)_4$	4)	$Sp^3 - Sp^3 - Sp^3$
		5)	$Sp^3 - Sp^2 - Sp^3$

A) $P-2, Q-5, R-3, S-4$ **B)** $P-2, Q-1, R-4, S-3$

C) $P-1, Q-2, R-4, S-3$ **D)** $P-1, Q-2, R-3, S-5$